



Wide area monitoring made simple

Hikvision's 'See Vaster' technology revolutionises area monitoring by integrating multiple lenses into a single camera. This replaces several conventional cameras, reducing costs and installation time. It features TandemVu and



How 'See Vaster' technology expands vision with multi-lens power
(Credit: <https://www.hikvision.com/en>)

PanoVu designs and simultaneously offers panoramic views and zoomed-in details. TandemVu's dual lenses provide a wide view and zoom functionality, while Smart Linkage tracks moving objects. AI features like facial recognition and ANPR deliver complete monitoring with reduced blind spots and lower maintenance.

Plastic supercapacitors could enhance energy storage

Researchers at UCLA have developed a new graphene-based material for supercapacitors, enhancing energy storage and performance. By using a vapor-phase growth process, they created vertical PEDOT nanofibres that significantly increase surface area and charge capacity. This material has 100 times the con-



Illustration of a PEDOT film on a graphene sheet that can be used in supercapacitors to store large amounts of energy (Credit: Maher El-Kady)

ductivity of traditional PEDOT and stores nearly 10 times more energy. It also lasts over 70,000 charge cycles, making it ideal for renewable energy applications and reducing dependence on fossil fuels.

Lithium-sulphur battery retains 80% capacity after 25,000 cycles

An international team of engineers and materials scientists from Chinese and German research teams believe they have made a breakthrough. They have developed a lithium-

sulphur battery that retains 80% charge capacity after 25,000 cycles, overcoming the limitations of current designs. Their breakthrough involves a novel sulphur-based electrode with a porous atomic structure enhanced by iodine to accelerate electron movement. This innovation enables rapid charging, high performance, and exceptional longevity. Despite the progress, researchers call for further advancements in energy density and material exploration for even more efficient, durable batteries.

New method found for creating soft electronics

Researchers at the National University of Singapore and Rice University have developed a new method for creating soft



Soft sensors engulfed on A4-size elastomer
(Credit: Rongzhou Lin.)

electronics using a phenomenon called 'particle engulfment.' This method embeds nanomaterials into a soft polymer matrix, overcoming challenges with traditional CNT dispersion. By combining particle engulfment

with a printing setup, they created stretchable, multi-layered electronics with wireless sensing, communication, and power capabilities. This breakthrough opens new possibilities for wearable devices, electronic skin, and soft robotics.

Human-like sensing by robots using artificial neurons

Researchers at Northwestern University and Georgia Tech have developed a high-performance organic electrochemical neuron (OECN) that mimics human neural frequency ranges, advancing artificial sensing. This neuron is part of a neuromorphic tactile perception system, integrating organic materials with touch receptors and synapses for real-time tactile signal processing. This achieves a



Human-like sensing by robots using artificial neurons (Credit: <https://www.mccormick.northwestern.edu>)

50-fold broader firing frequency range than existing circuits, enabling future robotics with enhanced sensory capabilities.